through POT. The linear displacement is connected to the sliding contact. The sliding contact is called as wiper. First, the wheatstone bridge is brought to balanced condition. At that time the galvanometer in the bridge shows null deflection.

Now the displacement is connected to the wiper. This linear displacement makes bridge to unbalance. The unbalance causes deflection in the galvanometer. This deflection is calibrated in terms of the applied displacement.

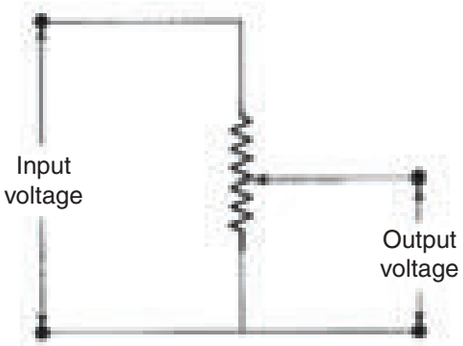


Fig. 7.3 Linear displacement

b. Potentiometers for rotary displacement

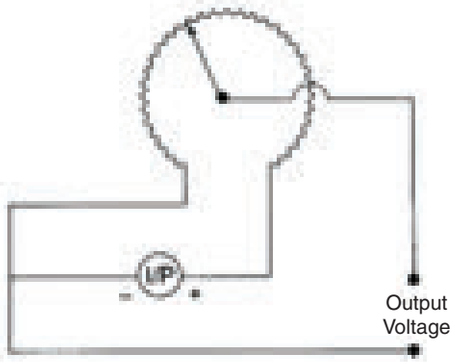


Fig. 7.4 Potentiometer for rotary pot

The input voltage is given to the two terminals of the potentiometer. The output voltage is taken from the output terminals as shown in figure 7.4. The slider arm is connected to the rotating quantity which we want measure. The rotating arm causes output voltage to change. This change in output is measured in terms of the applied rotary motion.

c. Strain gauge

The strain gauges are used to measurement of strain and associates stress. It works on the principle of piezo-resistivity. If a metal conductor is stretched or compressed, is resistance changes on account of the reason length and diameter changes.

d. Unbonded strain gauge

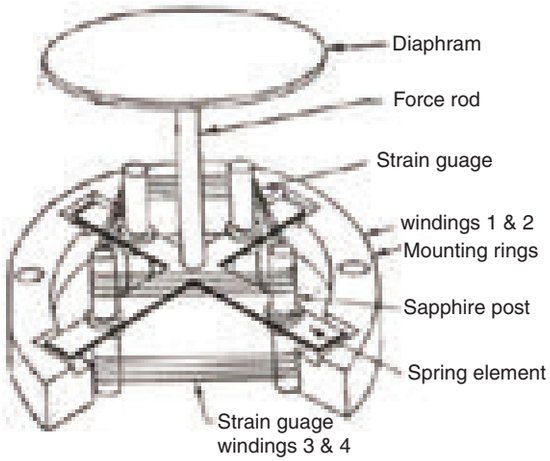


Fig. 7.5 Unbonded strain gauge

This gauge consists of a wire stretched between two points in an insulating medium such as air. The unbonded wires may be made up of various copper nickel, chrome nickel or nickel ion alloys.

The unbonded strain gauge wire acts as the four arms of a wheatstone bridge. Before applying pressure, the bridge is brought under balanced condition. Pressure is applied on the top of the diaphram as shown in figure 7.5. This makes unbonded wire to stretch along

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